Group Coursework Submission Form

Specialist Masters Programme

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MSc in: Financial Mathematics &Mathematical Trading and Finance	
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Lecturer: Gianluca Fusai	**Submission Date:** April 9, 2020

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0.1 SPAN System

EWMA model is usually applied when we implement scenario testing on our portfolio. As this approach do not assume equal weight for all historical observations when calculating the periodic volatility, rather, it assigns each observation with a weight that is time-decayed. Thus, old observations will have less influence in forecasting future data. Mathematically, it is a recursive formula

$$\sigma_n^2 = \lambda\sigma_{n-1}^2 + (1-\lambda)u_{n-1}^2 \tag{1}$$

where σ_{n-1}^2 is the volatility on day $n-1$ given by EWMA, u_{n-1}^2 is the squared reuturn on day $n-1$, and λ is the decay parameter. For daily data, $\alpha = 0.94$ provided by 'RiskMetrics' is usually adopted. However, to better fit our given S&P500 index data, we obtained our $\alpha = 0.90$ by minimising the squared volatility forecasting error. Given our fitted EWMA model

$$\sigma_n^2 = 0.90\sigma_{n-1}^2 + (1-0.90)u_{n-1}^2 \tag{2}$$

and the squared daily log-return on 31st Dec. 2019, we can forecast the daily volatility of S&P500 index return on next day at $\sigma_n = 0.4125\%$. Using this forecasted volatility, we can consruct 16 different market scenarios for our SPAN system, with 2 of them are extreme conditions that can be used for stress test. Those market conditions are specified by

$$\begin{aligned} &\text{S\&P500}_\text{t} e^{\pm j\cdot\hat{\sigma}_k\cdot\sqrt{10}}, \qquad k \in \{\text{Low}, \text{High}\} \\ &\text{S\&P500}_\text{t} e^{\pm 4\cdot\hat{\sigma}\cdot\sqrt{10}} \quad \text{(extreme conditions)} \\ &\hat{\sigma}_\text{Low} = \frac{\hat{\sigma}}{1.1} \quad \hat{\sigma}_\text{High} = 1.1\hat{\sigma} \end{aligned} \tag{3}$$

where we have $\text{S\&P500}_t = 3230.78$ at 31st Dec. 2019. According to (3), the 16 scenarios used in SPAN sytem can be obtained, as showed in Table 1.

j	$\hat{\sigma}_{Low} = 0.3750\%$	$\hat{\sigma} = 0.4125\%$	$\hat{\sigma}_{High} = 0.4537\%$
4		3403.82	
3	3347.78		3372.88
2	3308.32		3324.83
1	3269.32		3277.47
0	3230.78		3230.78
-1	3192.70		3184.76
-2	3155.06		3139.39
-3	3117.87		3094.66
-4		3066.54	

Table 1: S&P500 Index at 16 Different Scenarios

First consider the long index future position in our portfolio. The initial value of a index future contract is $\text{S\&P500}_t \times \$250 = \$807695.01$. According to Table 1, the two worst cases are S&P500 = 3094.66 and S&P500 = 3066.54, with weight 1 and 0.35 respectively. Thus, the worst loss on the long future position within 10 days will be -\$34029.07 which is also the initial margin for this position.

The value of index future options can be calculated analytically using Black Scholes,

$$\begin{aligned}
&\text{Call} = e^{-r(T-t)}(F\mathcal{N}(d_1) - K\mathcal{N}(d_2)) \\
&\text{Put} = e^{-r(T-t)}(K\mathcal{N}(-d_2) - K\mathcal{N}(-d_1)) \\
&d_1 = \frac{\ln(\frac{F}{K}) + (\frac{\sigma^2}{2})(T-t)}{\sigma\sqrt{T-t}} \\
&d_2 = d_1 - \sigma\sqrt{T-t}
\end{aligned} \tag{4}$$

where F is the index future. Since we have zero risk-free rate, which makes our spot S&P500 the same with its future contract. Based on the SPAN system reported in Table 1, the detailed SPAN report for each position in our portfolio is summarised in the following table, and the initial margins are calculated for each position separately.

Position	Initial Value	Worst Case Value	Extreme Case Value	Worst Loss	Initial Margin
Long 1 Future	\$807695.01	\$773665.94×1	\$766634.04×0.35	-\$34029.07	-\$34029.07
Short 2 Puts	\$329.46	\$595.48×1	\$651.58×0.35	-\$226.02	-\$226.02
Long 1 Call	\$323.33	\$191.64×1	\$163.95×0.35	-\$131.69	-\$131.69

Table 2: SPAN System for Our Positions

From Table 2, if netting is not allowed, which means we have to post full amount of initial margin for each position, then the total initial margin required is simply the sum of each margin, that is \$34386.78. In reality, netting is usally allowed if there are offsetting positions in the portfolio, as this can help to reduce the liquidity risk. However, in our case, we are essentially at a long S&P500 index position, and all of our positions are in the same direction, which indicates that loss will occur together and the diversification benefits do not exist. Thus, non-offsetting positions we held makes the initial margin with and without netting the same. This protects the clearing house, as initial margin will not be deducted when a risky portfolio is held.

0.2 Statistical Analysis of Stocks

Stock	Min	1st Qu.	Median	Mean	3rd Qu.	Max
INTC	-9.54%	-0.76%	0.094%	0.067%	0.86%	10.03%
JPM	-7.20%	-0.57%	0.059%	0.068%	0.752%	7.80%
AAPL	-10.49%	-0.587%	0.094%	0.094%	0.895%	7.88%
MSFT	-9.71%	-0.58%	0.083%	0.105%	0.804%	9.94%

Table 3: Summary Statistics form R

From Table 3, it can be found that all stocks in our portfolio have mean daily log-return around zero. Student-t test can be applied to test whether those stocks have zero daily return. We used a two-tailed t test with the null H_0: $\mu = 0$. According to Figure 2 , we can conclude that INTC's daily log-returns are insignificantly different from 0%, while JPM, AAPL and MSFT have daily log-returns that are significantly different from 0%, at 95% confidence level. This means our positions in those technology companies's stock can add value to our entire investment, if held for a long period. Based on the histograms in Figure 1, the daily log-returns are close to the Gaussian case. Assuming a Gaussian distribution for each stock's return allows us to do some

data analysis more straightforwardly, however, we should notice that, our empirical distributions are more skewed.

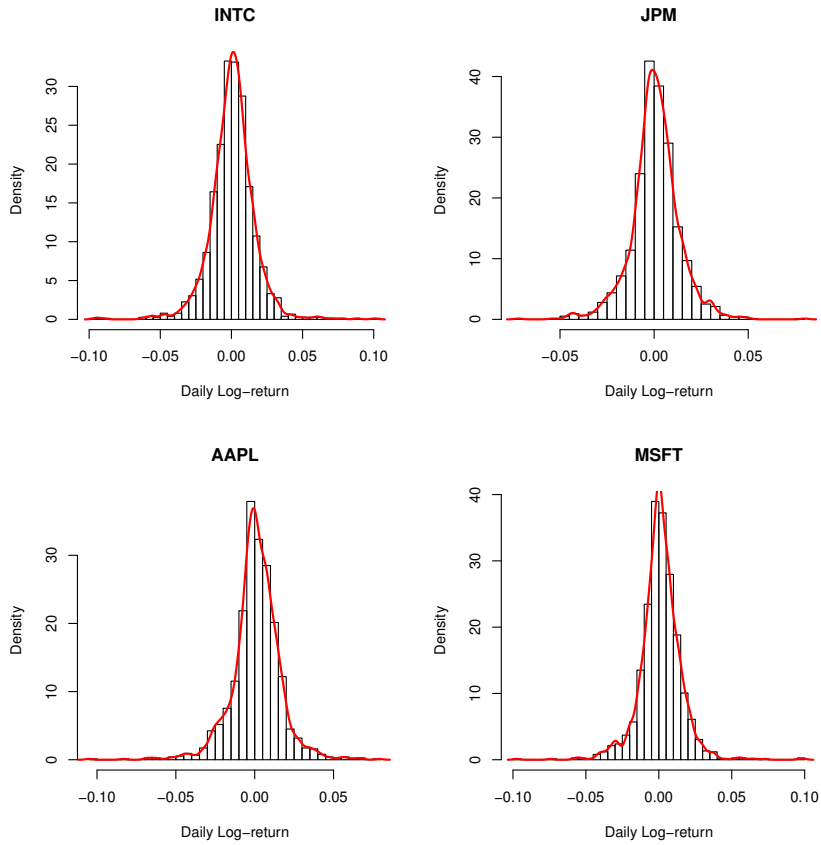
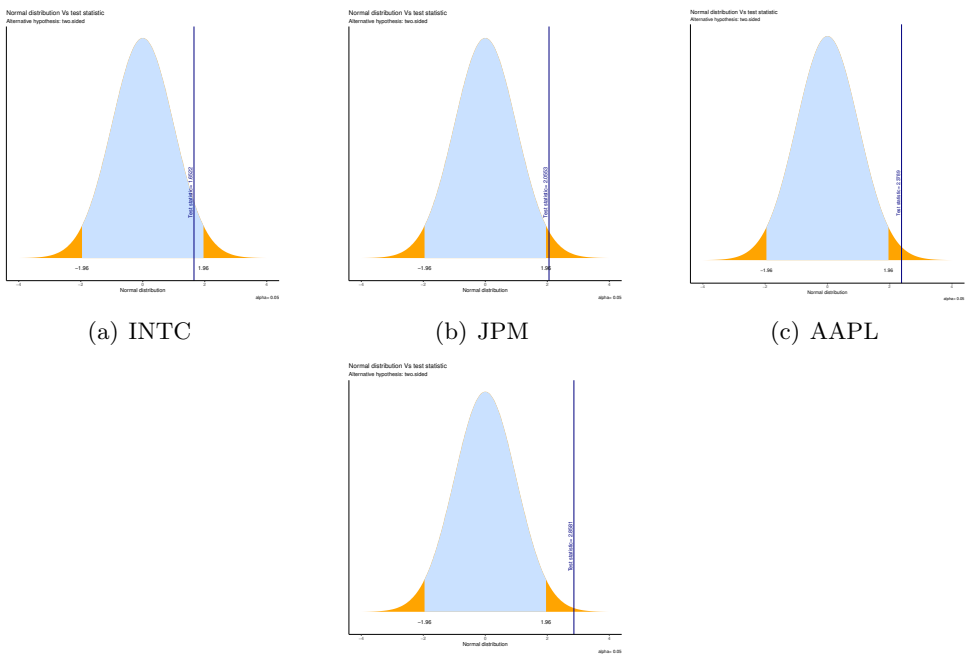


Figure 1: Histograms of Each Stock's Daily Log-return



(a) INTC (b) JPM (c) AAPL

(d) MSFT

Figure 2: Two-tailed Student-t Test Results

We first forecast the daily VaR at 90% and 99% confidence level, using the parametric Gaussian model, from July 1st. 2014. The estimated daily VaRs based on the data in our 6-month rolling window are showed in the below figure. It is clear that there are less violations under the 99% VaR model.

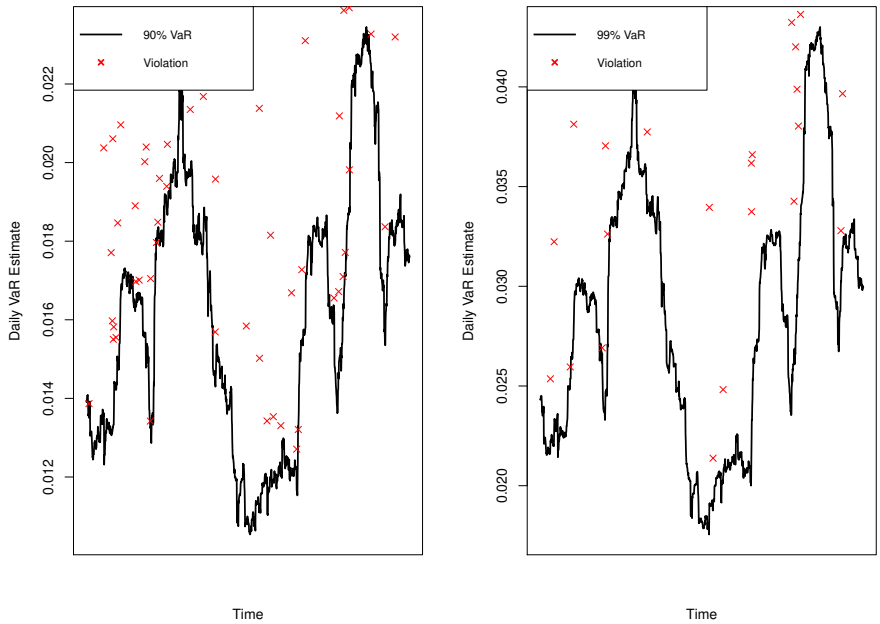


Figure 3: Daily VaR Estimates at 90% and 99% (Parametric Gaussian)

If the non-parametric historical simulation method is used, i.e. we simply forcecast the VaR based our historical information without any assumption on the distribution, then the model becomes

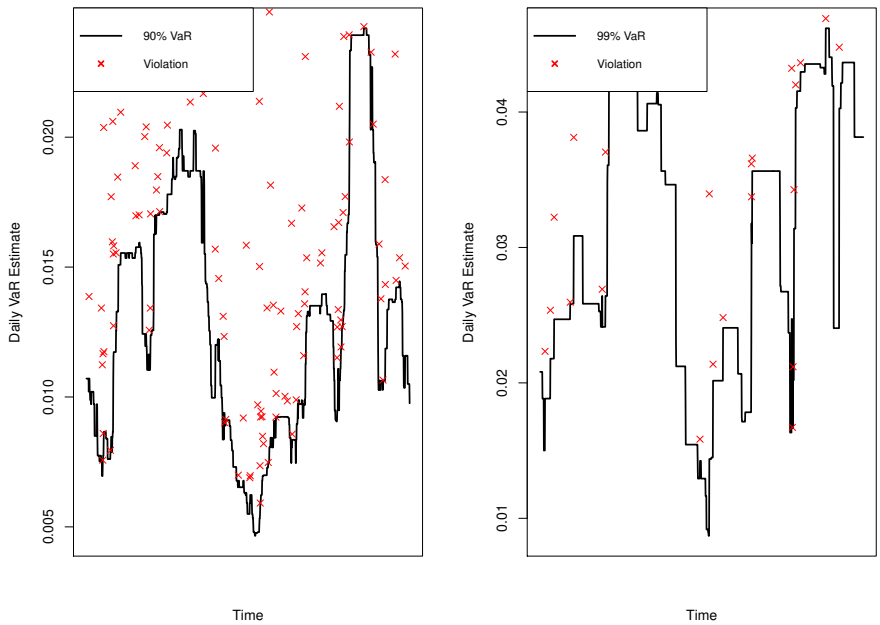


Figure 4: Daily VaR Estimates at 90% and 99% (Historical Simulation)

VaR Model	90% Level	99% Level
Parametric Gaussian	101	30
Historical Simulation	156	25

Table 4: Number of Violations in Each Model (Out of 1386)

To evaluate which model is more accurate, we can use an unconditional coverage Kupiec test which models the number of violations as a Bernoulli random variable. Asymptotically, the test statistic we use is

$$\mathrm{LR_{uc}} = -2\ln((\frac{1-\alpha}{1-\hat{\alpha}})^j(\frac{\alpha}{\hat{\alpha}})^{n-j}) \quad \sim \mathcal{X}_1^2 \tag{5}$$

with $\mathrm{H}_0: \ \hat{\alpha} = \alpha$, where α is the confidence level of our VaR measure, and $\hat{\alpha} = (1 - \frac{j}{n})$ which is the observed probability of not exceeding the estimated VaR in our dataset. We performed this test at confidence level of 95%, thus, we need to compare our $\mathrm{LR_{uc}}$ with the 95% quantile of $\mathcal{X}_1^2$ wich is about 3.84. Plugging in $n = 1386$, we can deduce that for a 90% VaR, the maximum numer of violation is 118 such that we will not reject H_0, and maximum 22 violations for a 99% VaR, in our sample size of 1386. 90% VaR model given by the parametric Gaussian is more close in this range, although this model seems under-estimate VaR at 99% confidence level.

VaR Model	90% Level	99% Level
Parametric Gaussian	Accurate	Inaccurate
Historical Simulation	Inaccurate	Inaccurate

Table 5: Accuracy Table According to Kupiec Test at 95% Confidence Level

However, Kupiec test can not be used as the only tool to assess our model accuracy. As the power of this test is low, which means this approach has a high probability to have a Type II error. Additionally, the underlying assumption of Binomial distributed violations may not valid in some case where the violations obviously show some serial dependency.

To show whether our violations in Figure 3 and Figure 4 under two models have serial dependency, the conditional coverage independent test devised by Christoffersen(1998) can be used. This is a statistical test of whether the VaR violations in the model have Markovian property, with test statistic constructed as

$$\begin{gathered}\mathrm{LR}_{cci} = -2\ln(\frac{(1-\pi)^{n_{00}+n_{10}}\ \pi^{n_{01}+n_{11}}}{(1-\pi_0)^{n_{00}}\ \pi_0^{n_{01}}\ (1-\pi_1)^{n_{10}}\ \pi_1^{n_{11}}} \quad \sim \mathcal{X}_1^2 \\ \text{where } \pi_0 = \frac{n_{01}}{n_{00}+n_{01}}, \quad \pi_1 = \frac{n_{11}}{n_{10}+n_{11}}, \quad \pi = \frac{n_{01}+n_{11}}{n_{00}+n_{01}+n_{10}+n_{11}}\end{gathered} \tag{6}$$

and n_{00} is the number of periods with no violation followed by no violation, n_{10} is the number of periods with violation followed by no violation, n_{01} is the number of periofd with no violation followed by a violation, n_{11} is the number of periods with violation followed by a violation. At 99% confidence level, we compare our test statistics with 6.6349, and the null hypothesis is that the VaR violations are not Markovian.

VaR Model	90% Level	99% Level
Parametric Gaussian	33.5257 (Reject)	5.71791 (Not Reject)
Historical Simulation	61.99544 (Reject)	2.450878 (Not Reject)

Table 6: Christoffersen's Interval Forecast Tests at 99% Confidence Level

According to the test results in Table 6, it can be seen that we cannot reject the null at 99% confidence level, which means the 99%VaR violations for both the parametric Gaussian and the historical simulation, are tend to be serially dependent.

Marginal VaR and component VaR for each asset in our portfolio can be easily calculated using our parametric Gaussian model. In Gaussian case, we have

$$\mathrm{MVaR}_\alpha = -\mu\Delta_n - \mathrm{z}_{1-\alpha}\frac{\sum_\Delta \mathbf{w}}{\sqrt{\mathbf{w}'\sum_\Delta \mathbf{w}}}\sqrt{n} \tag{7}$$

where μ is our mean return vector, Δ_n is the time horizon, $\sum$ is the variance-covariance matrix, and $\mathbf{w}$ is the weight vector. Once we obtained the MVaR matrix, The component VaR for each stock can be calculated by the MVaR multiplied by each stock's weight. Using matrix operation in R, we can get the results in Table 7.

Stock	MVaR	Weight	CVaR
INTC	2.229%	0.04438	0.09893%
JPM	2.046%	0.24074	0.49262%
AAPL	3.468%	0.58203	2.018329%
MSFT	2.4187%	0.13285	0.321314%

Table 7: 99% 1-day MVaR and CVaRs

MVaR is a useful indicator to better manage our portfolio VaR dynamically. To make our portfolio close to the efficient frontier, i.e. to maximise our risk-adjusted return, we can long more asset that has the lowest MVaR and short asset that has the highest MVaR. In our case, that is at current time, buying more JPM and selling AAPL. However, this strategy needs to be revised frequently, as the MVaR will change from time to time. In principle, we keep buying the lowest MVaR asset and selling the highest MVaR asset, we will eventually reach the most efficient point for our portfolio, and therfore, the Sharpe Ratio can be maximised.

From our Gaussian 99% 1-day VaR model, we can calculate the average of 10-day VaR over the previous 60 trading days as,

$$\frac{1}{60}\sum_{i=0}^{59}\mathrm{VaR}_{0.99}(t-\frac{i}{255}, t-\frac{i-10}{255}) = 9.947\% \tag{8}$$

and the 10-day $\mathrm{VaR}_{0.99}$ estimate at current time is $\mathrm{VaR}_{0.99}(t, t+\frac{10}{250}) = 9.269\%$. According to the capital regulation in Basel rule, the minimum capital requirement is calculated by

$$\mathrm{MCR(t)} = max(\mathrm{VaR}_{0.99}(t,\ t+\frac{10}{250}), S(t)\times\ \frac{1}{60}\sum_{i=0}^{59}\mathrm{VaR}_{0.99}(t-\frac{i}{255}, t-\frac{i-10}{255})) \tag{9}$$

where the current multipication factor $S(t)$ is based on Basel's traffic light system. The basic point of $S(t)$ is 3, and there will be 0.00 to 1.00 extra penalty according to the number of

violations of 1-day 99%VaR over last 250 days. Our Gaussian VaR model has total 4 violations, which leads to a 0.00 penalty factor and gives the final multiplication factor $S(t) = 3$. Thus, the minimum capital we need to hold for regulatory purpose will be

$$\begin{aligned} \text{MCR(t)} &= max(9.269\%, 3 \times \ 9.947\%) \times \Pi_t \\ &= max(9.269\%, 3 \times \ 9.947\%) \times \$9374.617 \\ &= \$2797.526 \end{aligned} \tag{10}$$

where Π_t is our current portfolio value at 31st Dec 2019.

0.3 Fitting Skewed Normal Distribution

The probability density function of a skewed normal distribution is

$$f(x) = \frac{2}{w}\phi(\frac{x-\xi}{w})\Phi(\alpha(\frac{x-\xi}{w})) \tag{11}$$

where ξ is the location parameter, w is the scale parameter, and α is the skewness parameter. $\phi(\cdot)$ and $\Phi(\cdot)$ denote the pfd and cdf of a standard normal distribution respectively.

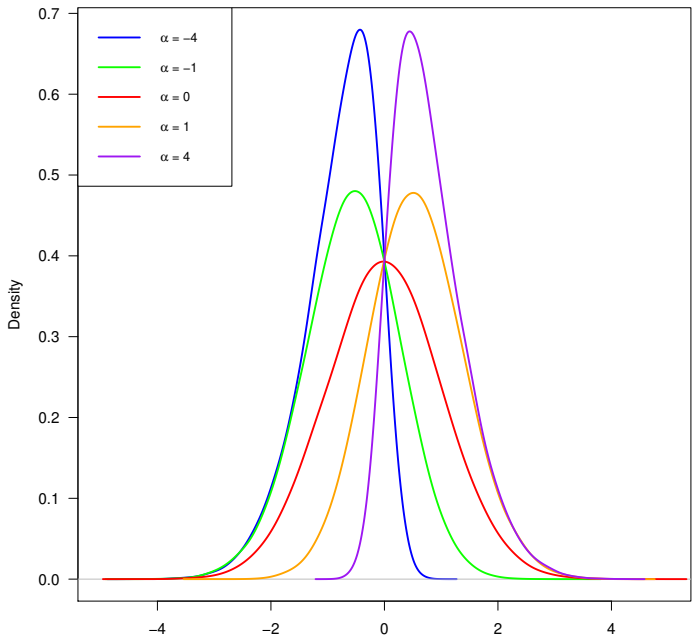


Figure 5: Density of Skewed Normal Distributions with different α

From Figure 5, if we vary the parameter α in the set $\{-4, \ -1, \ 0, \ 1, \ 4\}$ and keep all other parameters the same, it can be seen that the parameter α controls the skewness of the distribution. When $\alpha = \ -4$, the distribution is negatively skewed, as it has longer left tail. And when $\alpha = \ 4$, the distribution is positively skewed, as it has longer right tail.

Kurtosis is also influenced by the parameter α. For a skewed distribution with very long tail in either direction, for example, when $\alpha = \ -4$, the excess kurtosis of such a distribution will be higher than zero, i.e. leptokurtic. And the same is ture for a distribution governed by a very

positive α. The relationship between the parameter α and the skewness, the excess kurtosis measures, is visualised in the following figure.

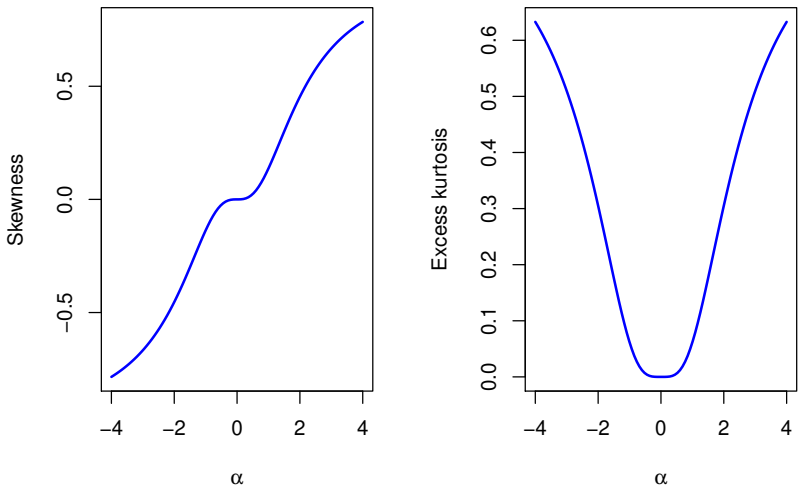


Figure 6: The Relationship Between α and The Skewness, The Excess Kurtosis

As an alternative choice of parametric distribution for financial data analysis, skewed normal distribution may fit better than the Gaussian for some skewed data. Figure 7 shows the empirical distribution of the daily log-return of APPLE stock (AAPL).

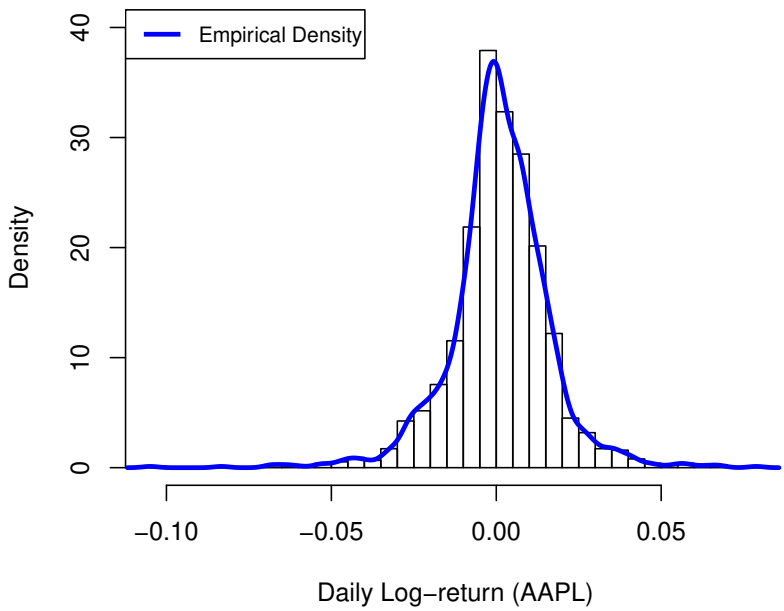


Figure 7: The Empirical Distribution of AAPL Daily Log-return

From the above figure, it can be seen that the empirical density of AAPL daily log-return is silightly skewed to the left, as its left tail is longer than the right. Thus, from Figure 5, such a

negatively skewed distribution should have a negative α, if fitted using the skewed normal distribution. The skewness can be calculated numerically using `skewness()` in R package `'moments'`, and for our AAPL dataset, we have `skewness(aapl_Ret)= -0.3794`, which is inline with our expecation of negative skewness based on Figure 5.

To fit the daily log-returns as Figure 7 shows using a skewed normal distribution, parameters in $\Theta \in \{w,\ \xi,\ \alpha\}$ need to be estimated properly. The Maximum-Likelihood estimation is usually used to estimated those parameters, by maximusing the log-likelihood function given each parameter. However, the closed-form solution of MLE for skewed normal distribution does not exist. In our implementation, method of moment is used to approximate the estimators given by MLE according to the following analytical formulas

$$
\begin{aligned}
|\hat{\delta}| &= \sqrt{\frac{\pi}{2}\frac{|\hat{\gamma_1}|^{\frac{2}{3}}}{|\hat{\gamma_1}|^{\frac{2}{3}} + (\frac{4-\pi}{2})^{\frac{2}{3}}}} \\
\hat{\alpha} &= \frac{\hat{\delta}}{\sqrt{1-\hat{\delta}^2}} \\
\hat{\sigma_x}^2 &= \hat{w}^2(1 - \frac{2\hat{\delta}^2}{\pi}) \\
\hat{\mu}_x &= \hat{\xi} + \hat{w}\hat{\delta}\sqrt{\frac{2}{\pi}}
\end{aligned}
\tag{12}
$$

where $\hat{\gamma}_1,\ \hat{\sigma}_x^2,\ \hat{\mu}_x$ are the sample skewness, sample variance and sample mean of our data respectively, and $\hat{\delta}$ has the same sign with $\hat{\gamma}_1$. Fitting our AAPL data, and solving the above MOM formulas, we obtained the following estimators

$$
\begin{aligned}
\hat{\delta} &= -0.8678 \\
\hat{\alpha} &= -1.7466 \\
\hat{w} &= 0.02127 \\
\hat{\xi} &= 0.01567
\end{aligned}
\tag{13}
$$

Using those estimated parameters, a fitted skewed normal density curve and be obtained.

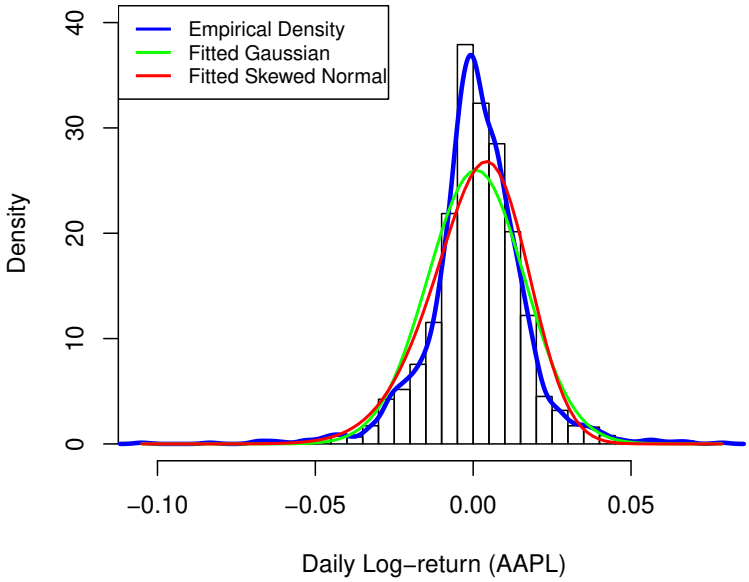


Figure 8: Empirical, Fitted Skewed Normal, and Gaussian Densities (AAPL Daily Log-return)

By fitting APPLE daily log-return to a skewed normal distribution using the parameters estimated in (3), the density curve of fitted skewed normal distribution as the red curve shows in Figure 8 can be plotted. However, as it can be found from the figure, the fitted skewed normal distribution behaves quite similar to the Gaussian distribution, except that the fitted skewed normal distribution is slightly skewed to left, and has a higher kurtosis than the Gaussian, which is more close to our empirical data. From the density plot above, it is difficult to decide which distribution fits the data better. Examining the fitness of two distributions by matching their theoretical percentiles with the percentiles from our empirical data, as Figure 9 shows, it is still inconclusive which one fits better, as both distributions matche the quantiles form our actual data well. However, it can be seen from Figure 9 that our fitted skewed normal distribution matches the lower quantiles better than the fitted Gaussian. This is a desired property as when evaluating the risk level of AAPL using VaR or ES which are related with lower quantiles, the fitted skewed normal can give more accurate risk estimation. And from Figure 8, the Gaussian has a thinner tail than the fitted skewed normal, which will underestimate risk levels at some extreme scenarios.

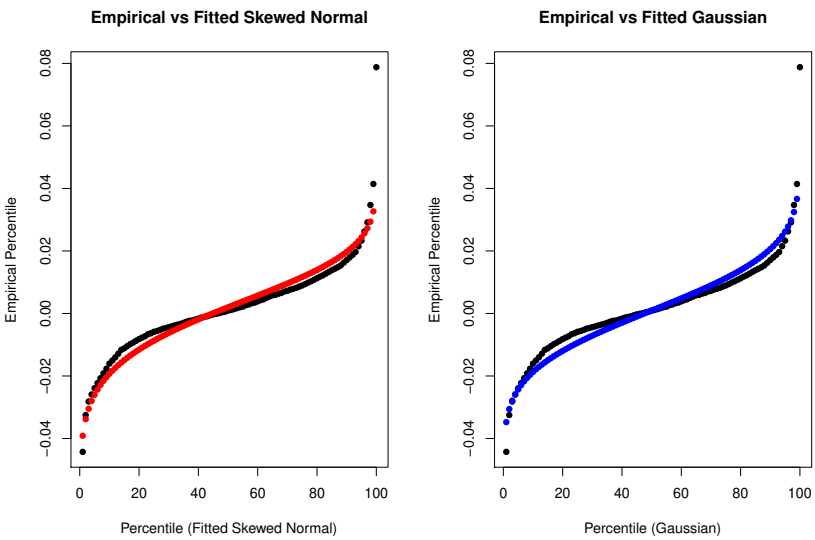


Figure 9: Percentile-to-Pencentile Matching of Fitted Skewed Normal and Fitted Gaussian

The pdf of our fitted skewed normal is

$$f(x) = \frac{2}{0.02127}\phi(\frac{x - 0.01567}{0.02127})\Phi(-1.7466(\frac{x - 0.01567}{0.02127})) \tag{14}$$

based on which further risk analysis using VaR and ES for AAPL can be done.

To examine the risk profile of AAPL, we will calculate its daily VaR at confidence level 90%, 95% and 99%. Since now we assume that the stock returns are skewed normally distributed, we can obtain those VaRs by calculating the corresponding quantiles under the skewed normal distribution with our parameters. Using `qsn()` function in R package `'sn'`, we obtained following quantiles in Table 8.

Significance Level (α)	0.01	0.05	0.1
Quantile (Fitted Skewed Normal)	-3.91%	-2.60%	-1.93%
Quantile (Fitted Gaussian)	-3.48%	-2.43%	-1.87%
Quantile (Empirical data/ HS)	-4.4268%	-2.3911%	-1.6053%

Table 8: 1%, 5%, 10% Quantiles

Thus, the 1-day VaRs of corresponding confidence level are $\text{VaR}_{99\%} = 3.91\%$, $\text{VaR}_{95\%} = 2.6\%$ and $\text{VaR}_{90\%} = 1.93\%$. From Figure 10, it is clear that our fitted skewed normal distribution gives more conservative VaR estimations than the Gaussian case. And at extreme cases, the fitted skewed normal fits our empirical data better, while the Gaussian distribution under-estimates VaR.

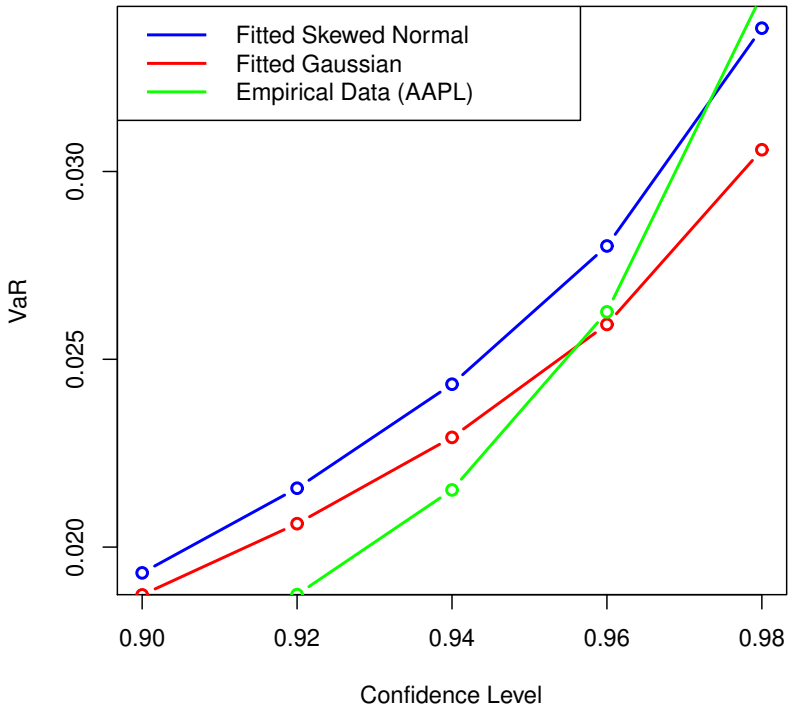


Figure 10: VaR as a function of $(1-\alpha)$ (Fitted Skewed Normal vs Gaussian)

Expected shortfall (ES) is defined as a tail confitional expectation,

$$\mathrm{ES}_{1-\alpha}(x) = \mathbb{E}(-x|x \leq -\mathrm{VaR}_{1-\alpha}) \tag{15}$$

which is the average loss when the VaR level has been exceeded. The ES at different confidence level under parametric and historical simulation approaches are summarised in the following table.

Significance Level (α)	0.01	0.05	0.1
ES (Fitted Skewed Normal)	4.58%	3.40%	2.82%
ES (Fitted Gaussian)	3.99%	3.07%	2.60%
ES (Historical Simulation)	5.938%	3.619%	2.809%

Table 9: 1%, 5%, 10% ES by Fitted Skewed Normal, Fitted Gaussian and HS approach

From above analysis, it can be seen that although the fitted Gaussian distribution and the fitted skewed normal distribution both fit our AAPL daily log-return well, and by conducting a Log-Likelihood Ratio test, they are insignificantly different in fitting our data, at 95% confidence level. However, because of the nature of our data, the fitted skewed normal distribution is preferred when analising AAPL daily log-returns under extreme confidence levels.

To simulate log-returns from our estimated skewed normal distribution, one simple method that is widely used in practice is 'Inverse transform sampling'. Since we know that the cdf of

our skewed normal distribution with parameters estimated in (3) is

$$\mathbb{P}(X \leq x) = F(X) = \Phi(\frac{x - \hat{\xi}}{w}) - 2T(\frac{x - \hat{\xi}}{w}, \hat{\alpha}) \tag{16}$$

where $T(\cdot)$ is the Owen's T function. By using inverse transfrom sampling, we need to generate the corresponding inverse cdf $F^{-1}(p) = x$. Owen's T function is available using the function `T.Owen()` in R package `'sn'`. The inverse cdf we obtained can also be considered as a quantile function of our fitted skewed normal distribution, which takes a probability as input, and returns the corresponding realisation of random variable x from our particular distribution.

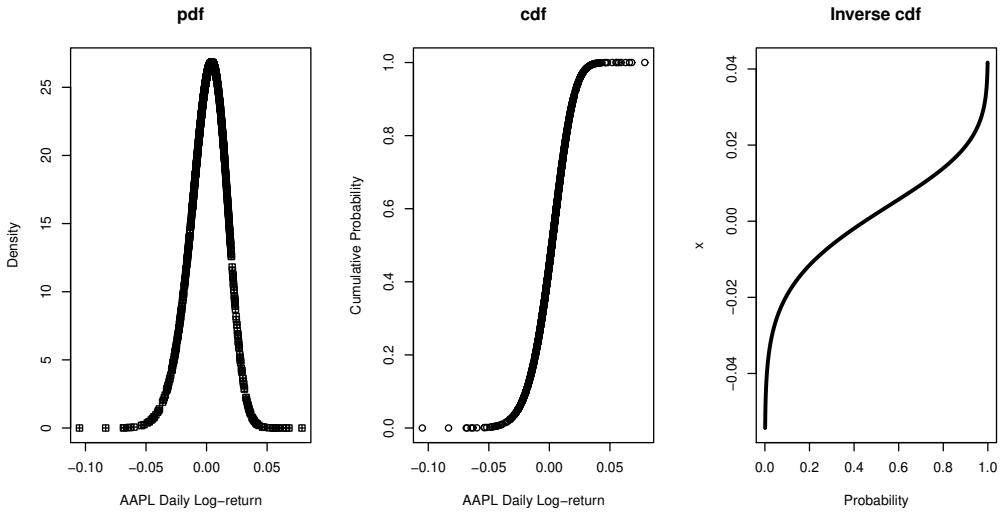


Figure 11: PDF, CDF and Inverse CDF of the Fitted Skewed Normal Distribution

As the required inputs for our inverse cdf, probabilities, are uniformly distributed, i.e. $p \sim \mathcal{U}(0, 1)$, and random numbers generated from a uniform distribution can be easily obtained in R, using `runif()`. Therefore, to simulate 20 i.i.d log-returns from our fitted distribution, we need first generate 20 uniformly distributed random numbers p. Plugging those random numbers p in our inverse cdf, we will then obtain 20 simulated i.i.d daily log-returns. Figure 12 belows shows the simulation result based on the above technique. From the density curve of our simulated 20-day log-returns, it can be found that they are negatively skewed, which indicates a negative α parameter, and is inline with our fitted skewed normal distribution. Thus, our simulated data are representative for our chosed distribution.

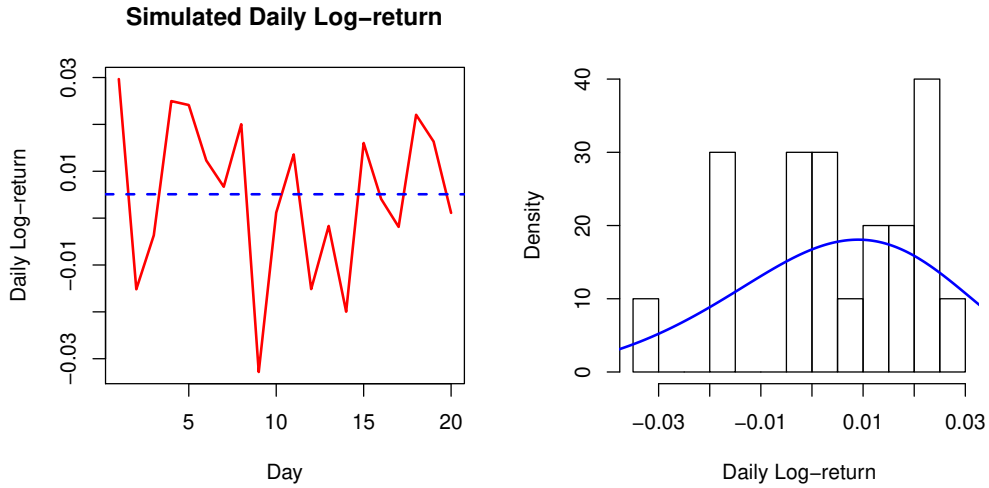


Figure 12: Simulated Data (Inverse CDF Method)

0.4 VaR of a Bond

0.4.1 Calculate the yield to maturity

The yield to maturity (YTM) of a bond is the rate when used to discount its cash flows, will give the present value of the bond. Therefore, in our case, the yield to maturity can be calculated using:

$$\$105 = \frac{\$2.5}{(1+\frac{\text{YTM}}{2})^1} + \frac{\$2.5}{(1+\frac{\text{YTM}}{2})^2} + \frac{\$2.5}{(1+\frac{\text{YTM}}{2})^3} + ... + \frac{\$102.5}{(1+\frac{\text{YTM}}{2})^{10}} \tag{17}$$

if semi-annual compounding is used. Solve (17) numerically, we get YTM = 3.89% for our 5-year coupon bond.

0.4.2 The probability losing more than 3% within 10 days:

Losing more than 3% in our bond position in 10 days means after 10 days, the price of bond will be $\$105 \times (1 - 0.03\%) = \$101.85$ and lower. Similar to formula (17), using the new price \$101.85, we can approximate a corresponding new $\text{YTM}^* = 4.582\%$ after 10 days. Since the bond price is negatively related to the yield to maturity, to lose more than %, the YTM needs to increase to 4.582% and higher, which means ΔYTM in 10 days should be larger than 0.6925%.

As we assume the absolute daily change in the YTM follows a Gaussian distribution i.e. $\Delta\text{YTM} \sim \mathcal{N}(0\%, (1\%)^2)$, and thus 10-day change in the YTM follows $\Delta^*\text{YTM} \sim \mathcal{N}(0\% \times 10, (\sqrt{10} \times 1\%)^2)$. We can then calculate the probability of $\Delta\text{YTM} \geq 0.6925\%$ as the follwing:

$$\begin{aligned}\mathbb{P}(\Delta^* YTM \geq 0.6825\%) &= \mathbb{P}(Z \geq \frac{0.6925\% - 0}{\sqrt{10} \times 1\%}) \\ &= 0.4972\end{aligned}$$

where $Z \sim \mathcal{N}(0, 1)$.

However, Monte Carlo simulation method can also be used to approximate the probability. Given the distribution of Δ^*YTM, we can simulate 100,000 scenarios of change of YTM in 10 days, and therefore, 100,000 scenarios of 10-day price change of our bond can be obtained. Figure 13 below illustrates the distribution of 10-day percentage return of our bond.

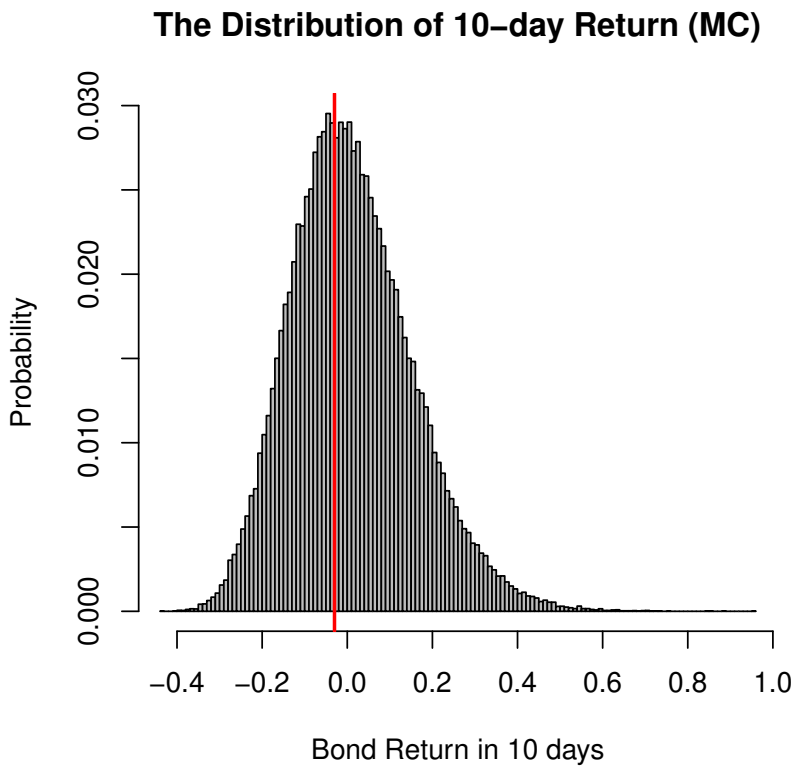


Figure 13: Monte Carlo Method to Calculate the Probability

The red line and its left shows scenarios where more than 3% loss in the bond position occurs, and the accumulative porbability is 0.4144. This result obtained from Monte Carlo simulation is close to the probability of 0.4972 by using exact formula, but the difference is still noticeable. More runs of simulation may reduce the difference between those two methods.

0.4.3 Exact Formula VaR

We know that the most important risk factor of our bond is the YTM, and if, for symplicity, we assume that the YTM is the only risk factor, then we can calculate the VaR based on the characteristic of the YTM. Using exact formula is one kind of parametric approach, as it is assumed that the underlying risk factor, in our case, the YTM, is Gaussian.

Recall that by the square-root-of-time rule, we have $\Delta^{10-day}\text{YTM} \sim \mathcal{N}(0\% \times 10, (\sqrt{10} \times 1\%)^2)$ for 10-day change. From formula (17), it can be seen that the value of bond is a decreasing funciton of the YTM, as YTM is in the denominator. Therefore, the best YTM level at the assigned horizon will lead to the worst case of bond price, at the given confidence level. The first step is then to calculate the best level of YTM, at 99% confidence level.

$$\begin{aligned} \text{YTM}_{\text{best}}\ (\text{t} + 10 \text{ days}) &= 0\% \times 10 + Z_{0.99} \times \sqrt{10} \times 1\% \\ &= 0 + 2.326 \times \sqrt{10} * 1\% \\ &= 7.3566\% \end{aligned} \tag{18}$$

According to (18), we know that the best Δ^{10-day} YTM level at 99% confidence level is 7.3566%, wich implies the best YTM will be $3.89\% + 7.3566\% = 11.2458\%$.

Once we have the best YTM, we can therefore calculate the worst bond price at 10-day horizon,

given the confidence level of 99%. Again, using formula (17), we will have the worst bond price at \$76.60. Thus, the 99% dollar VaR at 10-day horizon is: $|\$76.6 - \$105| = \$28.40$. And the corresponding percentage VaR is 27.0488%.

0.4.4 Delta Approximation VaR

Since the VaR measure is essentially the biggest change in price, i.e. $\Delta\ C(P(t),\ t)$, at a given confidence level (where C denotes the price level of the derivative, and P(t) denotes its major risk factor that we concern), we can use Taylor's expansion to approximate ΔC.

$$\Delta C(P(t),t) \approx \frac{\partial C(P(t),t)}{\partial t}\ dt + \frac{\partial C(P(t),t)}{\partial P}\ dP(t) + \frac{1}{2}\frac{\partial^2 C(P(t),t)}{\partial P^2}\ (dP(t))^2 \tag{19}$$

Above formula (19) is the Taylor's approximation up to the second order. For Delta-Normal approach we used in this section, the only first two terms will be considered. The first term is the partial derivative of the price respect to time, which is also knwon as Theta (Θ). It describes the sensitivity of price to time decay. To calculate Θ of our bond, we will use finite difference method. The forward difference used to approximate Θ is as the following:

$$\Theta \approx \frac{C(P(t),t+dt) - C(P,t)}{dt} \tag{20}$$

Keep the YTM = 3.89% at the begining unchaged, if we move one day forward, then we should use the following formula to calculate $C(P(t),t+dt)$:

$$\begin{aligned} C(P(t),t+dt) &= \frac{2.5\%}{(1+\frac{YTM}{2})^{1-\frac{1}{250}}} + \frac{2.5\%}{(1+\frac{YTM}{2})^{2-\frac{1}{250}}} + \frac{2.5\%}{(1+\frac{YTM}{2})^{3-\frac{1}{250}}} + ... + \frac{102.5\%}{(1+\frac{YTM}{2})^{10-\frac{1}{250}}} \\ &= 1.050111 \end{aligned}$$

Therefore, Θ of our bond is:

$$\begin{aligned} \Theta &\approx \frac{1.050111 - 1.05}{\frac{1}{250}} \\ &\approx 0.027750 \end{aligned}$$

The second term is the first derivative of bond price to the YTM, which is the bond's Delta (Δ). It can be approximated by central finite difference as formula (21) says:

$$\Delta \approx \frac{C(P+dP) - C(P-dP)}{2dP} \tag{21}$$

where $dP = 0.01\%$ which corresponding to a 1bp change in the YTM. Formula (21) gives us:

$$\begin{aligned} \Delta &\approx \frac{1.049567 - 1.050494}{2 \times \frac{0.01}{100}} \\ &\approx -4.63539 \end{aligned}$$

It can be found that the absolute value of the bond's $\Delta = |-4.63539|$ is close to its Modified duration of 4.41453. And Bond's Δ is sometimes substituted by negative Modified duration.

Having the bond's Θ and Δ in hand, we can then use Taylor's approximation in (19). Plug $\Theta = 0.02775$ and $\Delta = -4.63539$ in, we have:

$$\begin{aligned} \Delta C(P(t),t) &\approx \Theta dt + \Delta dP(t) \\ &\approx 0.02775 \times \frac{10}{250} + (-4.63539) \times 7.3566\% \\ &\approx -33.98952\% \end{aligned}$$

where $dP(t)$ is the 10-day increase in the YTM at 99% confidence level as in formula (18). Thus, by Delta approximation, we conclude the percentage 10-day VaR at 99% confidence level is 33.98952%, which equivalent to the dollar VaR of \$35.69. We notice that compared with the VaR of 27.0488% using exact formula in section 1.4.3, Delta approximation gives a larger estimate of VaR.

0.4.5 MC VaR with Delta-Gamma Approximation

As Delta approximation is just a linear approximation of the change in bond price, there will be substantial approximation error if the YTM increases or decreases a lot.

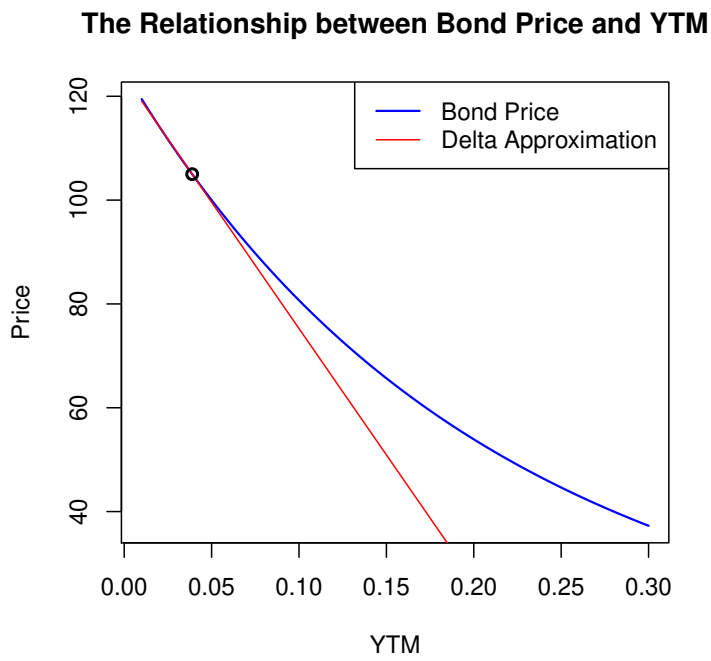


Figure 14: Delta Approxiamtion of Our Bond

As Figure 14 above illustrates, due to the curvature of bond price, Delta approximation deviates from the true price when the YTM changes a big amount. Moreover, particularly when the YTM increases, the change in bond price aprroximated by Delta approximation is large then its actual change, which over-estimates the risk level. This explains why we get a higher VaR in the previous section.

To adjust the bias in Delta approximation, the second order term, i.e. Gamma (Γ) term in Taylor's formula (19)) should be included, and it can also be obtained using a central finite difference:

$$\Gamma \approx \frac{C(P + dP_t) - 2C(P_t) + C(P - dP_t)}{(dP)^2}$$

where $dP_t = 1\text{bp}$ as well.

However, when actually using this finited difference method to obtain the second order derivative for our bond, the result will explode. This is because the price formula of our bond in (18) is poorly behaved, which will lead to finite difference method inaccurate. In this case, we need to change to other better numerical methods to approximate this Γ term. In application of finance industry, one widely used approximation of a bond's Γ is its convexity. As convexity

is a measure of the second derivative of the price of the bond with respect to the YTM.

$$\text{Convexity} = \frac{1}{\text{C(P(t), t)} \times (1+\text{YTM})^2} \sum_{\text{t=1}}^{5} \left[\frac{\text{CF}_{\frac{\text{t}}{2}}}{(1+\text{YTM})^{\frac{\text{t}}{2}}} ((\frac{\text{t}}{2})^2 + \frac{\text{t}}{2}) \right] \tag{22}$$

where CF_t is the cash flow of the bond at each time t. Using this formula, we can get the convexity of our bond which is 24.23491. And we use this as our approximate of Γ term in Taylor's expansion.

Having calculated our bond $\Theta = 0.027750$ $\Delta = -4.63539$ and $\Gamma = 24.23491$, we then plug those Greeks in Taylor's approximation and get:

$$\Delta C(P(t), t) \approx 0.027750 \times \frac{10}{250} + (-4.63539) \times dP(t) + \frac{1}{2} \times 24.23491 \times (dP(t))^2 \tag{23}$$

Different from read off the best change of YTM from its distribution as we did in Delta approximation, Monte Carlo simulation evaluate $\Delta C(P(t), t)$ based on the larg number of simulations on the YTM. Run 100,000 simulations, we will get 100,000 simulated change of our bond price.

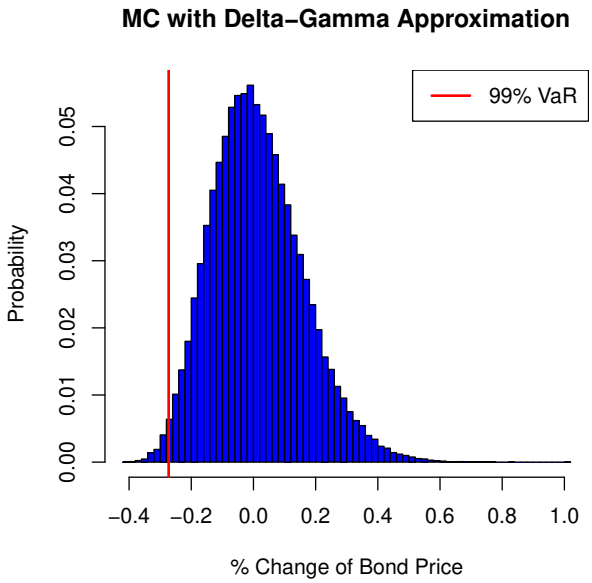


Figure 15: 100,000 runs of MC Simulation with Delta-Gamma Approximation

The simulated distribution of percentage change of our bond in 10 days is in Figure 15. We can calculate the 1% quantile of this distribution, which is -27.2753%. This is equivalent to 99% VaR of 27.2753%. As we include the convexity adjustment in Delta-Gamma approximation, the upward bias in Delta approximation can be corrected in some extent. This is inline with our empirical result, as our Delta-Gamma-approximated VaR is lower than Delta-approximated VaR.

0.4.6 MC VaR with Full Revaluation

Alternatively, if more accurate estimate of VaR is required, Monte Carlo with full revaluation can be used. As this approach does not use Taylor's approximation to approximate changes in the position, but re-calculate the position value by its exact pricing formula. However, this approach is still a parametric approach, as some parametric distribution such as Guanssian is assumed on the risk factor.

According to our assumption on the 10-day change of YTM, we can simulate 100,000 new YTMs, and then calculate 100,000 new bond prices. By doing this, we can obtain a distribution of percentage returns in 10 days, as Figure 16 below shows.

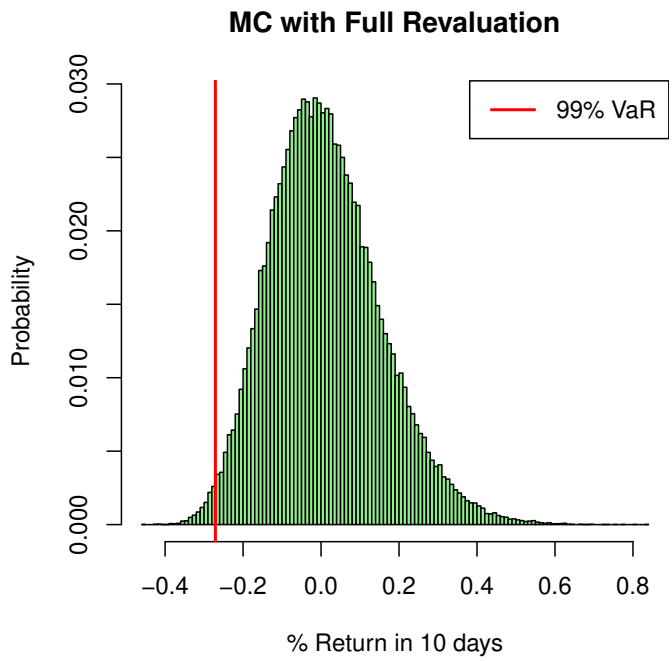


Figure 16: 100,000 runs of MC Simulations wit Full Revaluation

This simulated distribution gives a 1% quantile of -27.1078%, which is equivalent to a 10-day 99% VaR of 27.1078%. This is result is close enough to the result from Monte Carlo with Delta-Gamma approxiamtion, which again testify that including Gamma term in Taylor's approximation helps to improve the accuracy of the VaR estimate.

0.5 VaR of An Option

Calculations in this section are based on the 1-Year USD LIBOR rate on 23-Mar-2020, which is 0.9685%.

0.5.1 The probability that the option price falls more than 10% within 20 days

To estimate the probability of losing more than 10% in our call option position within 20-day period, we need first calculate what will be the minimum dollar amount loss during that period. Using the colsed form Black-Schole's formula, we know that the ATM call option price with the annualised volatility $\sigma = 30\%$, 1 year to maturity, the risk-free rate $r = 0.9685\%$, and spot price of the underlying $S_0 = \$100$, at time 0 is $C_0 = \$12.3541$. Thus, more than 10% loss means there should be more than $C_0 \times 10\% = \$1.2354$ dollar term loss. Again, according to Black-Schole's formula, we can approximate the option delta Δ_C by taking its first order partial derivative with respect to the underlying

$$\begin{aligned} \Delta_C &= \frac{\partial \mathrm{C_t}\ (\mathrm{S_t},\ \mathrm{K},\ \mathrm{r},\ \sigma,\ \mathrm{T-t})}{\partial S_t} \\ &= \mathcal{N}(d_1) \end{aligned} \tag{24}$$

where $d_1 = \frac{\log(\frac{\mathrm{S_t}}{\mathrm{K}})+(\mathrm{r}-\frac{1}{2}\sigma^2)(\mathrm{T}-\mathrm{t})}{\sigma\sqrt{\mathrm{T}-\mathrm{t}}}$. Plug in our parameters, we can therefore get $\Delta_C = 0.5723$ for our call option.

By the definition of option Delta, we have the following relationship

$$\begin{aligned} \Delta_C &= \frac{dC}{dS} \\ \Delta_C \cdot dS &= dC \end{aligned} \tag{25}$$

Thus, to have a more than \$1.2354 loss over 20 days in the call option position, $dS = \$2.1586$ within 20 days in our case. Therefore

$$\begin{aligned} \mathbb{P}(dC \leq -\$1.2354) &= \mathbb{P}(dS \leq -\$2.1586) \\ &= \mathbb{P}\left\{\log(\frac{S_0 + dS}{S_0}) \leq -2.1822\%\right\} \\ &= \mathbb{P}(r_{20} \leq -2.1882\%) \end{aligned} \tag{26}$$

As the the daily stock log-returns are i.i.d Gaussian, i.e.

$$r \sim \mathcal{N}(0, (\frac{1}{\sqrt{250}} \times 30\%)^2) \tag{27}$$

we will therefore have $r_{20} \sim \mathcal{N}(0, (\frac{\sqrt{20}}{\sqrt{250}} \times 30\%)^2)$, which is the distribution of 20-day log-returns, if serially uncorrelated of log-returns on each day is assumed. Thus, according to the parametric distribution of r_{20}, we have

$$\mathbb{P}(r_{20} \leq -2.1882\%) = 0.3985$$

which means the probability that the call price will fall more than 10% within a 20-day period is 0.3985.

0.5.2 Exact Formula VaR

Using exact formula to compute VaR, the best or worst level of the underlying risk factor at a given confidence level needs to be determined first, according to the parametric distribution the risk factor. In our case, the underlying risk factor is the stock price. As the daily stock log-returns are Gaussian with $\mu = 0$ and $\sigma = 30\%$, the stock price will be log-normally distributed. Since the call option price is an increasing function of the risk factor, to compute VaRs , we need to determine what is the worst level of the stock price. As we are interested in the $\mathrm{VaR}_{99\%}$ within 1, 10, 20, 30, ..., 90 days, we need to calculate the worst stock price level in those periods respectively. By the square-root-of-time rule, we can obtain the distributions of stock log-returns within each of given horizons. And according to (27)

$$\begin{aligned} &S_{\text{worst}}(t+\Delta t) = S_t e^{0\cdot\Delta t + Z_{0.01}\sigma\frac{\sqrt{\Delta t}}{\sqrt{250}}} \\ &\text{where } \Delta t = \begin{bmatrix} 1 & 10 & 20 & 30 & 40 & 50 & 60 & 70 & 80 & 90 \end{bmatrix} \end{aligned} \tag{28}$$

we can now obtain a worst stock price vector $\mathbf{S_{worst}}$ for the horizons we are interested in. Then the final step is pricing the call option with $\mathbf{S_{worst}}$ and new time to maturities, using Blakc-Scholes formula. Comparing the new call option prices at Δt with the original price $C_0 = \$12.3541$, we can therefore obtain dollar-term $\mathrm{VaR}_{99\%}$ for the desired horizons, as Table 10 shows.

Δt	1	10	20	30	40	50	60	70	80	90
$\mathrm{VaR}_{99\%}$	\$2.37	\$6.49	\$8.38	\$9.53	\$10.33	\$10.90	\$11.32	\$11.62	\$11.85	\$12.02

Table 10: $\mathrm{VaR}_{99\%}$ at Different Horizons Using Exact Formula

0.5.3 Delta Approximation VaR

The option VaR which essentially is the maximum loss of its price over a given period at a certain confidence level. Taylor's expansion can be used to approximate such a change $\Delta C(S_t, t)$ in the option price. The Taylor's expansion up to the first-order, i.e. Delta approximation is

$$\Delta C(S_t, t) = \frac{\partial C(S_t, t)}{\partial t} dt + \frac{\partial C(S_t, t)}{\partial S_t} dS_t \tag{29}$$

where in (29), the first partial derivative term is the option Theta (Θ), and the second partial derivative term is the option Delta (Δ). By finite difference method, we have

$$\begin{aligned} \Theta &\simeq \frac{C(S_0, 0 + \frac{1}{250}) - C(S_0, 0 + \frac{1}{250})}{\frac{1}{250}} \\ &\simeq -6.3263 \\ \Delta &\simeq \frac{C(S_0 + 1, 0) - C(S_0 - 1, 0)}{2 \times 1} \\ &\simeq 0.5723 \end{aligned} \tag{30}$$

if we assume $dt = \frac{1}{250}$ and $dS = 1$. Using the vector of worst stock price at 99% confidence level $\mathbf{S_{worst}}$ derived from (12), we can calculate $d\mathbf{S_t}$ by $\mathbf{S_{worst}} - S_0$, for our given horizons. And the actual dt is given by $d\mathbf{t} = \frac{1}{250}\Delta t$. Therefore, by Delta approximation using exact formula, we have the following VaRs showed in Table 11.

Δt	1	10	20	30	40	50	60	70	80	90
$\text{VaR}_{99\%}$	\$2.50	\$7.71	\$10.76	\$13.05	\$14.95	\$16.61	\$18.09	\$19.44	\$20.69	\$21.86

Table 11: $\text{VaR}_{99\%}$ at Different Horizons Using Exact Formula via Delta Approximation

Compare the VaRs in Table 11 with the VaRs in Table 10, it can be found that Delta approximation gives a higher estimate of VaR at each horizon. According to Figure 17, the Black-Scholes (purple surface) shows some convexity, which makes the Delta approximation (red surface) return a higher loss when the stock price falls, and a lower profit when the stock price increases. This explaines why the VaRs given by Delta approximation are generally larger than as exact formula gives in Table 10. From Table 10 and Table 11, this discrepancy between two methods increases when the time horizon widens, as longer horizon increases the volatility of log-returns which will therefore lead to a lower worst stock price. Additionally, Delta approximation via exact formula gives some unreasonable large VaRs that have already exceeded the initial option price C_t.

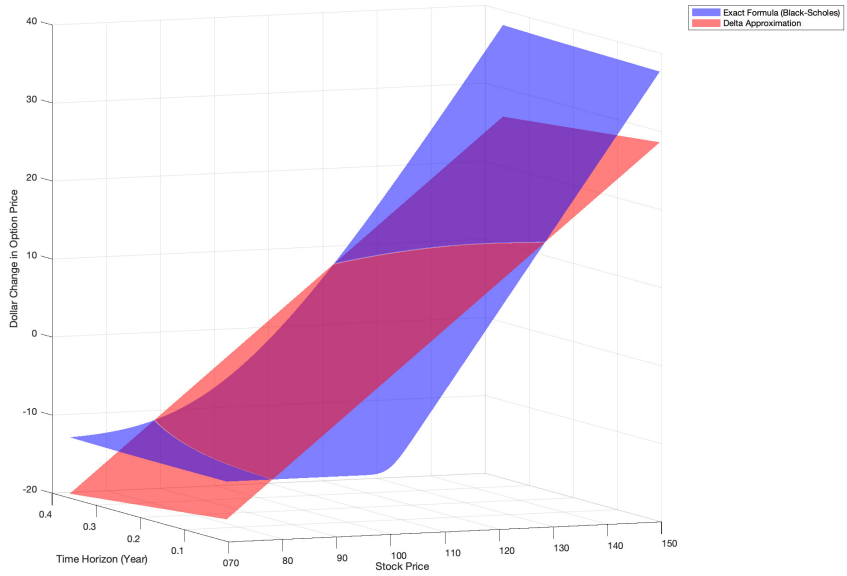


Figure 17: Change of Option Price Using Black-Scholes and Linear Approximation

0.5.4 Delta-Gamma Approximation VaR

Such a discrepancy can be narrowed if the second-order term, i.e. Gamma term is included in our Taylor's approximation, as (31) describes

$$\Delta C(S_t, t) = \frac{\partial C(S_t, t)}{\partial t} dt + \frac{\partial C(S_t, t)}{\partial S_t} dS_t + \frac{1}{2} \frac{\partial^2 C(S_t, t)}{\partial S_t^2} (dS_t)^2 \tag{31}$$

And the Gamma term can be approximated by the following finite difference if we let $dS = 1$

$$\begin{aligned} \Gamma &\simeq \frac{C(S_t + 1, t) - 2C(S_t, t) + C(S_t - 1, t)}{1^2} \\ &\simeq 0.01307822 \end{aligned} \tag{32}$$

Use again the worst stock price change vector $d\mathbf{S} = \mathbf{S_{worst}} - S_0$, and the time horizon vector $d\mathbf{t}$, the Delta-Gamma approximated VaR via exact formula can be calculated as Table 12 shows

Δt	1	10	20	30	40	50	60	70	80	90
$\text{VaR}_{99\%}$	\$2.37	\$6.60	\$8.66	\$10.03	\$11.07	\$11.91	\$12.61	\$13.21	\$13.73	\$14.20

Table 12: $\text{VaR}_{99\%}$ at Different Horizons Using Exact Formula (Delta-Gamma Approximation)

From the above table, it can be found that, compared with Delta approximation, Delta-Gamma approach gives VaR estimations that are more close to the result in Table 10 given by exact formula than Delta approximation. This is because that by adding the Gamma term, the downward bias in Delta approximation when the stock price falls as showed in Figure 17 can be reduced.

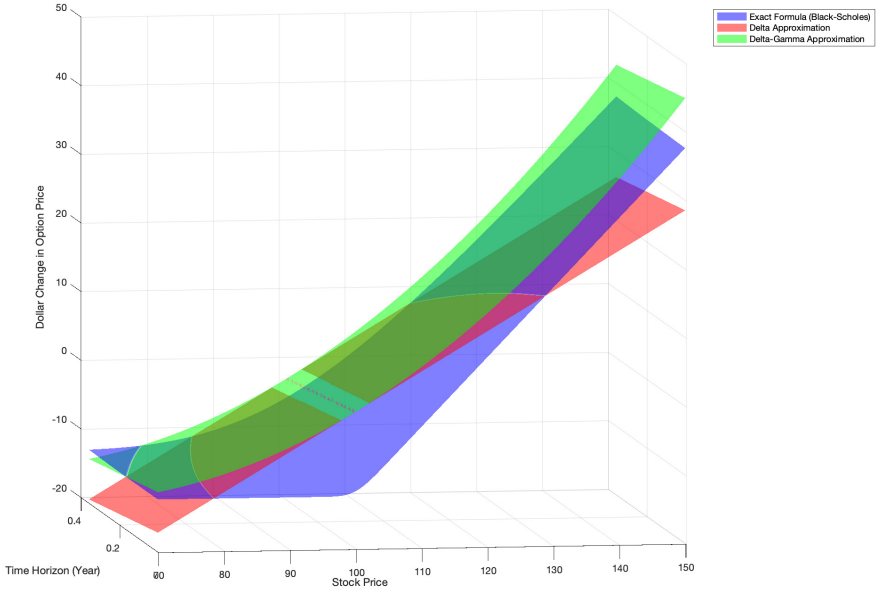


Figure 18: Explicit Formula, Delta Approximation and Delta-Gamma Approximation

As Figure 18 shows, when the stock price falls, Delta-Gamma approximation returns a lower loss than Delta approximation, but is higher than the Black-Scholes surface. The surface given by Delta-Gamma approximation is much more closer to the exact formula at low stock price level and high stock price level. Thus, in Table 12, the VaRs given by Delta-Gamma approximation are more accurate than the VaRs in Table 11, and are in a reasonable range. However, for those longer horizons such as 70-day, 80-day and 90-day, it still gives a VaR that exceeds our initial option value, which can be avoided if better approximation method is used.

Although both Delta and Delta-Gamma approximation are not accurate enough, as (29) and (31) shows, they consider the time decay effect (Theta-Effect) in calculating the call option value change. Therefore, from Figure 18, it can be found for those two approximation methods, option value will decrease more when the time horizon increases, which is in line with the risk management of options in practice as time risk needs also to be considered, while exact formula only captures the sensitivity of option value to the stock price.

0.5.5 MC VaR with Delta Approximation

Monte Carlo simulation can be used with Tyalor's approximation, to better estimate our option VaR. This approach does not require us to calculate the worst $d\mathbf{S}$ at 99% confidence level as Delta and Delta-Gamma approximation. The major different of this approach with previous approaches is that, because of Monte Carlo, the $d\mathbf{S}$ will be obtained using simulation method.

$$dS_t = S_t(e^r - 1) \tag{33}$$

According to (33), as the distribution of log-returns r is known, dS_t can be simulated by simulating r. Thus, a large number of P&L of our option can be simulated according to the Taylor's approximation. The distributions of simulated option P&L over our given time horizons are in Figure 19.

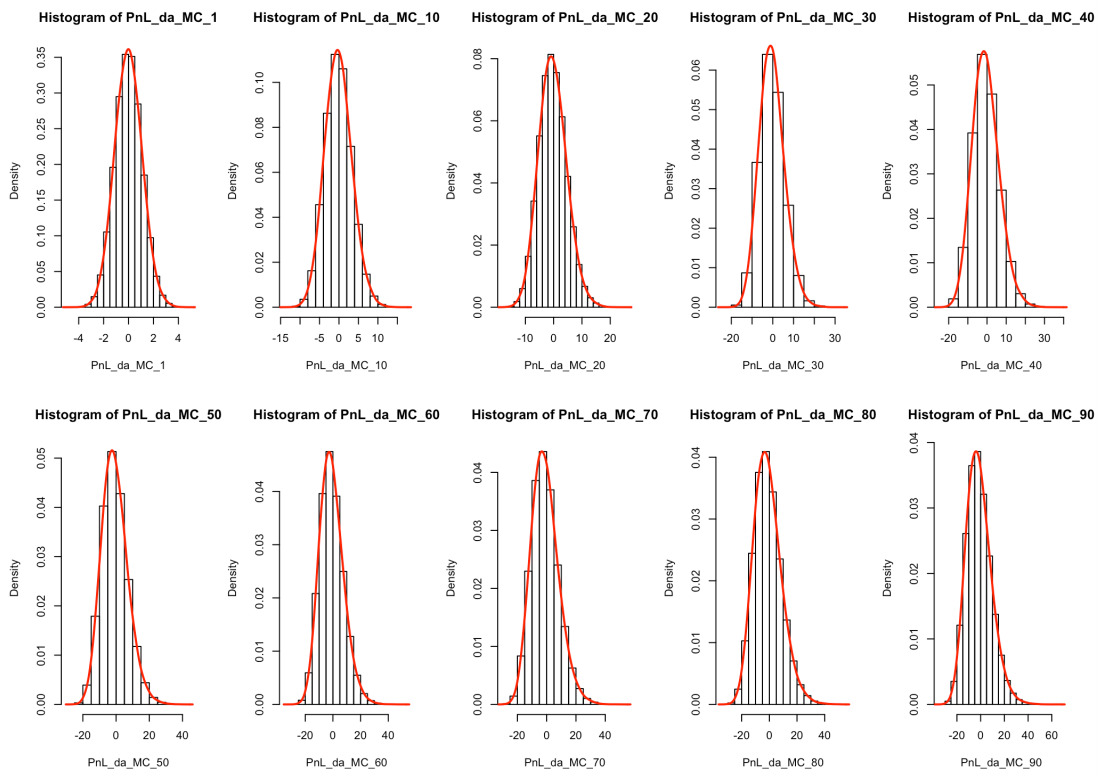


Figure 19: Option P&L using MC with Delta Approximation (100,000 Runs)

Therefore, the 1% quantiles that give our $\text{VaR}_{99\%}$ estimations can be obtained from those distributions

Δt	1	10	20	30	40	50	60	70	80	90
$\text{VaR}_{99\%}$	\$2.49	\$7.74	\$10.78	\$13.05	\$14.96	\$16.72	\$18.19	\$19.39	\$20.72	\$21.88

Table 13: $\text{VaR}_{99\%}$ at Different Horizons Using Monte Carlo Simulation (Delta Approximation)

Again, due to the downward bias of Delta approximation mentioned previously, MC with Delta approximation still gives over-estimated VaRs, as MC simulation only changes the randomness in dS_t.

0.5.6 MC VaR with Delta-Gamma Approximation

Follow the same procedure of Delta-Gamma approximation in (31), but substitute the MC simulated dS_t and $(dS_t)^2$. The following distributions of option P&L as Figure 20 shows can be obtained. Compare the P&L distributions given by Delta approximation in Figure 19 with the distributions by Delta-Gamma approximation in Figure 20, it is obvious that including a quadratic Gamma term alters the distibution of option P&L. Such a regime shift becomes more obvious in longer horizons.

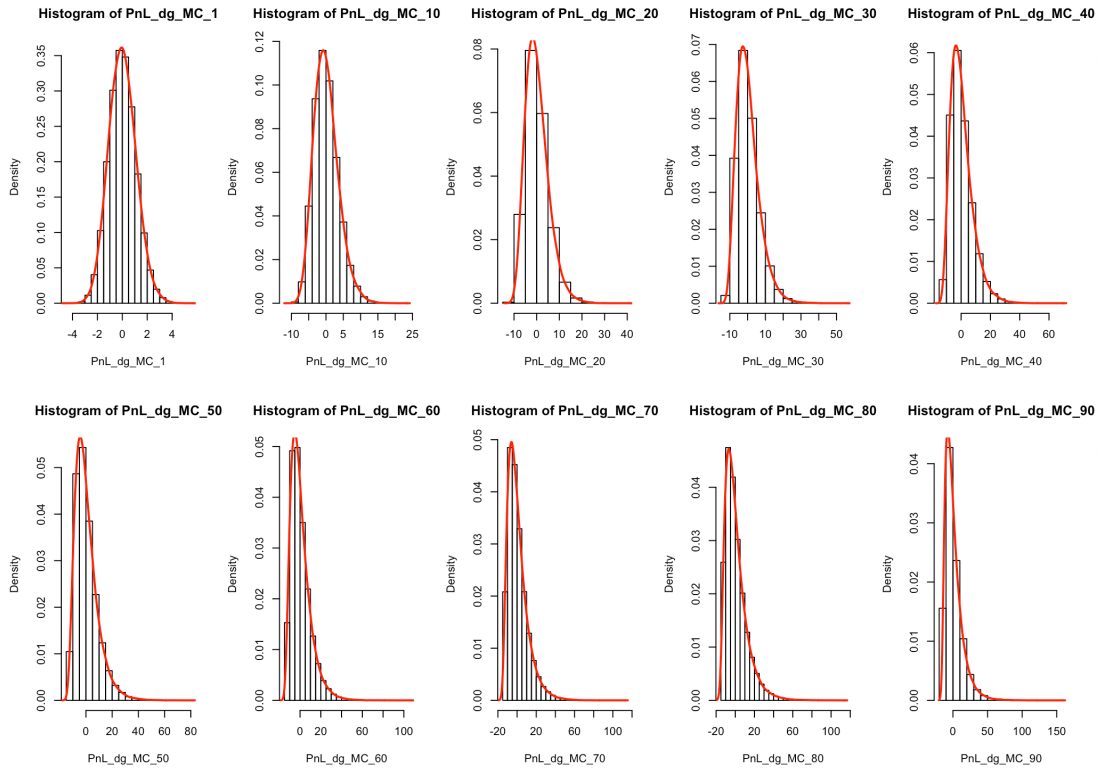


Figure 20: Option P&L using MC with Delta-Gamma Approximation (100,000 Runs)

$\text{VaR}_{99\%}$ obtained from above distributions are summarised in Table 14 below

Δt	1	10	20	30	40	50	60	70	80	90
$\text{VaR}_{99\%}$	\$2.37	\$6.62	\$8.67	\$10.04	\$11.08	\$11.95	\$12.64	\$13.19	\$13.74	\$14.21

Table 14: $\text{VaR}_{99\%}$ Using Monte Carlo Simulation via Delta-Gamma Approximation

The MC results are similar to the results by Delta-Gamma approximation with exact formula. And because of the advantage of including a Gamma term, over-estimated VaRs by Delta approximation can be corrected to some extend. However, the limitaions due to Taylor's approximation approach such as it may lose accuracy when the stock price falls dramatically, and it can give a VaR that exceeds the initial value still exist.

0.5.7 MC VaR with Full Revaluation

Taylor's approximation approach discussed above can give a numerical answer quickly. However, to get rid of the bias introduced by Taylor's approximation, Monte Carlo simulation with full revaluation can be used. Different from Taylor's approximation, this approach does not approximate the change of option value by its Greeks. We directly simulate a large number of stock prices at each horizon, according to the given distribution of log-returns. And then, closed form formulas such as Black-Scholes are used to calculate the option value given our simulated stock prices. Thus, a distribution of MC simulated option P&L can be obtained, from wich the $\text{VaR}_{99\&}$ can be calculated. Table 15 shows our results from MC with full revaluation.

Δt	1	10	20	30	40	50	60	70	80	90
$\text{VaR}_{99\%}$	\$2.36	\$6.52	\$8.39	\$9.54	\$10.33	\$10.93	\$11.34	\$11.61	\$11.85	\$12.02

Table 15: $\text{VaR}_{99\%}$ Using Monte Carlo Simulation via Full Revaluation

VaRs given by full revaluation MC are very close to the VaRs by exact formula in Table 1. This is because full revaluation is done on each simulated scenario based on the analytical formula we choose. Thus, unlike Taylor's approximation approach, this method has less loss of accuracy. And more importantly, by using the analytical formula, the minimum option value will be bounded at 0, which is more consistent with the reality. Thus, full revaluation retuns VaRs that are all below the intial value of our call option, as $C_0 = \$12.35$ is the maximum we can lose on a long call position.

0.6 Marginal VaR in The Gaussian Case

Proof. If returns are multivariate Gaussian, then the computaion of marginal VaR using

$$\mathbb{E}(r_i | r_P = -\text{VaR}) \tag{34}$$

agrees with the explicit formula of MVaR in the Gaussian case.

If we assume that in a portfolio, asset log-returns are multivariate Gaussian with mean vector $\vec{0}$ and variance-covariance matrix Σ, i.e.

$$\vec{r} \sim \mathcal{N}(\vec{0}, \Sigma) \tag{35}$$

The mean vector $\vec{r}$ of an n-assets portfolio is:

$$\begin{bmatrix} r_1 \\ r_2 \\ r_3 \\ \vdots \\ r_n \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

And the variance-covariance matrix Σ of an n-assets portfolio is:

$$\Sigma = \begin{bmatrix} \sigma_1^2 & \sigma_{1,2} & \cdots & \sigma_{1,n} \\ \sigma_{2,1} & \sigma_2^2 & \cdots & \sigma_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{n,1} & \sigma_{n,2} & \cdots & \sigma_n^n \end{bmatrix}$$

Then, the conditional distribution of log-return r_i of asset i given the portfolio log-return r_p is also Gaussian and $r_i | r_p \sim \mathcal{N}(\mu_{i,p}, \Sigma_{ii,p})$, where:

$$\begin{aligned} \mu_{i,p} &:= \mathbb{E}(r_i | r_p) = \mu_i + \Sigma_{ip}\Sigma_{pp}^{-1}(r_p - \mu_p) \\ &= \mathbb{E}(r_i | r_p = -VaR) \\ &= r_i + \Sigma_{i,p}\Sigma p, p^{-1}(-VaR - r_p) \\ &= 0 + \sigma_{i,p}(\sigma_p^2)^{-1}(-VaR - 0) \\ &= -\sigma_{i,p}(\frac{1}{\sigma_p^2})VaR \end{aligned} \tag{36}$$

The above holds if we assume the portfolio log-return r_p is equal to its portfolio $VaR_p = -VaR$ and condition r_i on r_p. According to the definition of Normal VaR, we have $VaR_p = 0 - z_\alpha \sigma_p = z_\alpha \sigma_p = -VaR$. And if we substitute $VaR = -z_\alpha \sigma_P$ into formula (36) derived above, we can get:

$$\begin{aligned} \mu_{i,p} &= -\sigma_{i,p}(\frac{1}{\sigma_p^2})(-z_\alpha \sigma_p) \\ &= \frac{\sigma_{i,p}}{\sigma_p} z_\alpha \end{aligned} \tag{37}$$

Moreover, the partial derivative of the portfolio variance σ_p^2 with respect to the weight w_i of asset i is:

$$\begin{aligned} \frac{\partial \sigma_p^2}{\partial w_i} &= \frac{\partial \vec{w}^\mathsf{T} \Sigma \vec{w}}{\partial w_i} \\ &= 2w_i\sigma_i^2 + 2\sum_{j=1, j\neq i}^{n} w_j \sigma_{ij} \\ &= 2(w_i\sigma_i^2 + \sum_{j=1, j\neq i}^{n} w_j \sigma_{ij}) \\ &= 2Cov(r_i, w_i r_i + \sum_{j\neq i}^{n} w_j r_j) \\ &= 2Cov(r_i, rp) \\ &= \sigma_{i,p} \end{aligned} \tag{38}$$

If we use chain rule to $\frac{\partial \sigma_p^2}{\partial w_i}$, we will also have $\frac{\partial \sigma_p^2}{\partial w_i} = 2\sigma_p \frac{\partial \sigma_p}{\partial w_i}$. And thus,

$$\begin{aligned} 2Cov(r_i, r_p) &= 2\sigma_p \frac{\partial \sigma_p}{\partial w_i} \\ \implies \frac{\partial \sigma_p}{\partial w_i} &= \frac{Cov(r_i, r_p)}{\sigma_p} = \frac{\sigma_{ip}}{\sigma_p} \end{aligned} \tag{39}$$

Recall that from the definition of Marginal VaR of asset i in the portfolio:

$$MVaR_i = \frac{\partial VaR_p}{\partial w_i} = \frac{\partial z_\alpha \sigma_p}{\partial w_i} = z_\alpha \frac{\partial \sigma_p}{\partial w_i} \tag{40}$$

Then combine (40) with (39), we have an alternative formular for $MVaR_i$:

$$MVaR_i = z_\alpha \frac{\sigma_{ip}}{\sigma_p} \tag{41}$$

Compare formula (41) derived from the definition of MVaR in multivariate Gaussian case with formula (37), it can be found that they are equivalent. As $\mu_{i,p}$ in formula (37) is the conditional expectation of r_i given $r_p = -VaR$, we then conclude that if returns are multivariate Gaussian, the computaion of MVaR according to $\mathbb{E}(r_i | r_p = -VaR)$ agrees with the explicit MVaR formula. □